News API & Web Search Latency Benchmarking — Report

This report provides a comparative analysis of system latency across two experimental runs conducted on **March 31, 2026**. The experiments evaluated the performance of specialized news APIs (**MarketAux**, **The Guardian**, **NewsAPI**) and two web search providers (**SerpAPI** and **Serper.dev**) integrated via LangChain community wrappers.

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1. Comparative Latency Overview

The following table summarizes the **mean latency** for each API across both test runs. Note that while the news-specific APIs were called in both runs, the web search provider was toggled between Serper.dev and SerpAPI.

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2. Provider Analysis

MarketAux: The Primary Bottleneck

In both experiments, MarketAux was the **"long pole"** in the system latency chain:

API / Tool	Intent Category	Serper Run (Mean)	SerpAPI Run (Mean)	Delta
MarketAux	Business News	8,627.30 ms	4,952.65 ms	−3,674.65 ms
Web Search	Web Search	1,532.73 ms	115.81 ms	−1,416.92 ms
The Guardian	Sports News	591.19 ms	486.68 ms	−104.51 ms
NewsAPI	General News	120.61 ms	139.03 ms	+18.42 ms

- **Peak Latency:** Recorded a maximum of **15,906.94 ms** (nearly 16 seconds) for a currency exchange rate query.
- **Variability:** The mean latency fluctuated significantly between runs (from **4.9 s** to **8.6 s**), suggesting the API is highly sensitive to network jitter or complex keyword processing.

Web Search: Serper vs. SerpAPI

Based strictly on the provided experimental data, the web search results showed a surprising reversal of typical industry expectations:

- **SerpAPI** outperformed **Serper** in these specific tests, maintaining a highly consistent **mean of 115.81 ms** and a **p95 of only 121.32 ms**.
- **Serper.dev** showed higher latency and greater variance, with a **mean of 1,532.73 ms** and a **p95 of 2,570.28 ms**.
- **Observation:** The extreme speed of SerpAPI in Run 2 (**~104 ms** for a sourdough bread query) suggests potential caching or an exceptionally optimized connection for that session.

Specialized News APIs

- **NewsAPI:** Consistently the fastest tool in the suite, frequently responding in **under 150 ms**.
- **The Guardian:** Demonstrated very stable performance around the **500–600 ms** mark across all calls.

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3. System Architecture Impact

The total elapsed time for the test suite highlights the cost of sequential execution:

- **Serper Run Total: 50.47** seconds.
- **SerpAPI Run Total: 27.48** seconds.

The "Long Pole" Conclusion: Because the testing script runs calls **sequentially**, a single slow response from MarketAux (**15.9 s**) can stall the entire pipeline. In a production LangGraph environment, **parallel fan-out** is critical. By running these tools in parallel nodes, the user's total wait time would be reduced from the **sum** of all latencies to only the **single slowest** call (the MarketAux response).

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4. Final Recommendation

1. **Tool Selection:** While SerpAPI was faster in this specific run, Serper remains a strong contender for web search if SerpAPI's latency regresses to its typical multi-second mean.
2. **MarketAux Optimization:** Given that MarketAux frequently exceeds **5 seconds**, you may want to implement a timeout or a parallel backstop to Web Search if the business agent takes too long.
   Other sources should be investigated. One choice mentioned in the original note was: `https://www.alphavantage.co/support/#api-key`
   Follow-up (post-report): Alpha Vantage (News & Sentiments) was evaluated; **strict request-rate limits** (e.g. free-tier throughput) make it a poor fit for interactive multi-call benchmarks and were a factor in **not** adopting it for the Business feed in this project.
3. **Graph Logic:** Use **NewsAPI (General News)** and **The Guardian (Sports)** results as high-confidence, low-latency components that can populate a UI nearly instantly, while the Business agent "streams" in later.

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5. Takeaway — Architecture update (latency-driven)

We are coalescing on NewsAPI for both Business and World (same **`/v2/everything`** integration), with Business vs World distinguished in the **supervisor / intent** layer—not via a separate slow Business feed.

Why this is consistent with the data above:

- **NewsAPI** was the **fastest** wired news API in these runs.
- **MarketAux** was the **dominant long pole** for Business-style queries; keeping it on the critical path works against responsive UX when calls are sequential.
- Routing **Business** and **World** through **NewsAPI** matches **`docs/requirements.txt`** and preserves **The Guardian** for **Sports**, where latency was already acceptable and stable.

Net: Latency findings support treating **NewsAPI** as the shared wire feed for **Business + World**, **Guardian** for **Sports**, and **web search** for categories outside that set (per product requirements).
